

SUPPLEMENTARY MATERIAL FOR REAL ANALYSIS (2026)

Pritam Chandra

The material in this document will supplement the lectures by the instructor. Although this document will mostly be self-contained, from time to time the knowledge of definitions and concepts covered in the class will be assumed.

1	Defining the real numbers	1
1.1	Ordered fields	2
1.2	Least upper bounds	3
1.2.1	Exercises	5
2	Sequences	6
2.1	Convergent and Cauchy sequences	6
2.2	Limit superior and limit inferior	10
2.2.1	Exercises	12
3	Series	13
3.1	Convergence, absolute and conditional convergence	13
3.1.1	Exercises	16

1 Defining the real numbers

Question 1.1. What is the difference between an *operation* and a *relation*? Think of examples.

A *group* G is a set equipped with the binary operation $+$ (addition) such that

1. the operation $+$ is associative, that is, for all $a, b, c \in G$, we have $a + (b + c) = (a + b) + c$,
2. there is an (identity) element $0 \in G$ such that $a + 0 = a$ for all $a \in G$,
3. for every $a \in G$, there is an element $-a \in G$ such that $a + (-a) = 0$.

By definition, if G is a group then $|G| \geq 1$. There is no empty group.

A *field* F is a set equipped with binary operations $+$ and $\cdot$ such that

1. $(F, +)$ is an Abelian group (the identity is 0)

2. $(F - \{0\}, \cdot)$ is also an Abelian group (the identity is 1)
3. $\cdot$ distributes over $+$, that is, $a(b + c) = ab + ac$ for all $a, b, c \in F$.

Question 1.2. Why is there no field of cardinality 1?

Again, by definition it is clear that if F is a field then $|F| \geq 2$. A field has at least an additive and a multiplicative identity which are distinct. However, in many other equivalent formulations of the field definition the condition $|F| \geq 2$ is explicitly incorporated. In such formulations, the distinctness of the additive and multiplicative identity is inferred. In fact, if $0 = 1$, then the field necessarily collapses to the singleton $\{0\}$. To see this, note that in F

$$0a = 0 \quad \forall a. \quad [\text{Proof. } 0a = (0 + 0)a = 0a + 0a \implies 0a = 0]$$

As a corollary, we get that if $0 = 1$, then $F = \{0\}$. In F we also have

$$(-1)a = -a \quad \forall a \quad [\text{Proof. } (-1 + 1)a = 0 \implies (-1)a = -a],$$

a small result which we will use in the next subsection.

1.1 Ordered fields

An *ordered field* is a field F equipped with a **total** order such that the field operations are compatible with the order in the following manner:

1. if $a \leq b$ then $a + c \leq b + c$ for all c ,
2. if $a \leq b$ and $c \geq 0$ then $ac \leq bc$.

So, we have seen that given a set, one can give structure to it in multiple ways. One way is to assign operations on the set. In this direction we get groups, fields etc. Another is to give it an order, which gives rise to ordered sets. An ordered field combines both the structures. The following diagram displays this as a hierarchy diagram.

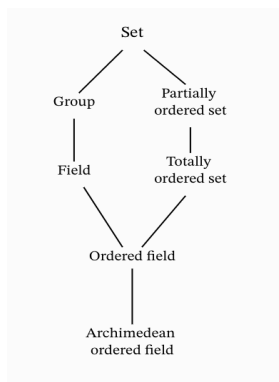


Figure 1.1. Hierarchy diagram of structured sets

We say an ordered field F is *Archimedean* if whenever $x, y \in F$ such that $x < y$, then there exists $n \in \mathbb{N}$ such that $y < nx$. Here, in a general field by nx we mean x added to itself n times.

Informally, the Archimedean property says that no element of the field is either too large or too small. Some properties of ordered fields are listed below with their proofs.

$a \leq b \implies -b \leq -a$	$a + (-b - a) \leq b + (-b - a) \implies -b \leq -a.$
$a \leq b \implies bc \leq ac \quad \forall c \leq 0$	$a \leq b \implies a(-c) \leq b(-c) \implies -ac \leq -bc \implies bc \leq ac.$
$a \geq 0, b \geq 0 \implies ab \geq 0$	$a \geq 0 \implies ab \geq 0b \implies ab \geq 0.$
$a^2 \geq 0 \quad \forall a$	Only need to show for when $a \leq 0$. In that case $-a \geq 0$. Then $a^2 = (-1)(-1)a^2 = (-a)(-a) \geq 0$.
$0 \leq 1$	$(-1)^2 = 1$ Incorrect proof: pick $a \geq 0$ such that $a \neq 0$. Then $0 \geq 1 \implies 0 \geq a$. Incorrect because such a choice of a may not exist.
$a \geq 0 \implies a^{-1} \geq 0$	If $a^{-1} \leq 0$ then $a^{-1}a \leq 0a \implies 1 \leq 0$. Contradiction.
$0 \leq a \leq b \implies 0 \leq b^{-1} \leq a^{-1}$	$a, b \geq 0$ then $a^{-1}b^{-1} \geq 0$. Multiply $a \leq b$ with $a^{-1}b^{-1}$.

Note that F being totally ordered is important in the proof of the statement $a^2 \geq 0$, and (therefore) for all the statements that follow from it.

1.2 Least upper bounds

An element $a \in S \subset W$ is said to be an *upper bound* for S if $a \geq s$ for all $s \in S$. Now, a is called a *least upper bound* or an *infimum* for S if whenever b is another upper bound for S , it holds that $a \leq b$. If S has an infimum it is unique. Therefore we denote this infimum as $\inf S$.

A poset W is said to have the *least upper bound (LUB) property* if whenever a subset S has an upper bound in W , it has a least upper bound in W . The natural analogues of these quantities are *lower bounds*, *greatest lower bounds or supremums*, and the *greatest lower bound (GLB) property*. The notation for the supremum of the set S is $\sup S$.

The set $\mathbb{Q}$ of rationals does not have the LUB property, while $\mathbb{Z}$ and $\mathbb{N}$ do. In fact, $\mathbb{Z}$ and $\mathbb{N}$ have something stronger. Any non-empty subset S of $\mathbb{Z}$ or $\mathbb{N}$, that has an upper bound in S , has its supremum in S itself. In order to show this we assume the following characterization of the integers.

Proposition 1.1. $\mathbb{Z}$ is the smallest subring of $\mathbb{R}$ whose positive elements, say $\mathbb{Z}_+$, are well-ordered, that is, if $S \subset \mathbb{Z}_+$ and S is non-empty, then S has a minimum.

The above also gives a characterization of the natural numbers as being the positive elements of $\mathbb{Z}$.

Corollary 1.1. Suppose $s, t \in \mathbb{Z}$. Then, either $s \leq t$ or $s \geq t + 1$.

Solution. Suppose not, that is, suppose $s, t \in \mathbb{Z}$ such that $t < s < t + 1$. Consider $r = s - t$. We have $r \in \mathbb{Z}$ and $0 < r < 1$. Let

$$S = \{r, r^2, \dots\}.$$

As $r > 0$, we have $r^n > 0$ for all $n \in \mathbb{N}$. Therefore, $S \subset \mathbb{Z}_+$. By the well-ordering principle, S has a minimum, say r^k . Further, as $r < 1$, we get $r^k r < r^k$ or $r^{k+1} < r^k$. This is a contradiction to the minimality of r^k . $\square$

Proposition 1.2. Let $S \subset \mathbb{Z}$ be non-empty such that S has an upper bound. Then $\sup S \in S$.

Solution. We know that $\sup S \in \mathbb{R}$. Call this number α . Note that $\alpha - 1$ is not an upper bound for S , therefore there exists $s \in S$ such that

$$s \leq \alpha \quad \text{and} \quad \alpha - 1 < s.$$

Let $t \in S$. Suppose $t \geq s + 1$. Then $\alpha < s + 1 \leq t$, which is a contradiction, as α is an upper bound for S . Therefore, we have $t \leq s$. This shows that s is an upper bound for S , which in turn implies that $\alpha \leq s$. Since $\alpha \geq s$ also, we conclude that $\alpha = s$. $\square$

Definition 1.1. The set of real numbers, denoted by $\mathbb{R}$, is the smallest ordered field containing $\mathbb{Q}$ (as an ordered field) that has the LUB property.

Using just this definition alone we can prove many familiar properties of $\mathbb{R}$. For $x \in \mathbb{R}$ let

$$U(x) = \{z \in \mathbb{Z} \mid z \geq x\} \quad \text{and} \quad L(x) = \{z \in \mathbb{Z} \mid z \leq x\}.$$

Since $\mathbb{Z}$ has the least upper bound property, we can define the following $\mathbb{Z}$ -valued functions on $\mathbb{R}$,

$$\lceil x \rceil = \inf U(x) \quad \text{and} \quad \lfloor x \rfloor = \sup L(x).$$

These are the familiar *ceiling* and *floor* functions, respectively. Below we list some properties of these functions.

$\lfloor x \rfloor \leq x \leq \lceil x \rceil$	To see this, note that x is a lower bound for $U(x)$. So $\lceil x \rceil = \inf U(x) \geq x.$ The other side of the inequality can be similarly argued.
$0 \leq \lceil x \rceil - \lfloor x \rfloor \leq 1$	The left inequality follows from point 1. To see the right one, note that the integer $\lfloor x \rfloor + 1$ is not in $L(x)$, as $\lfloor x \rfloor$ is an upper bound for $L(x)$. Since $\mathbb{Z}$ has a total order, it must be that $\lfloor x \rfloor + 1 \geq x$. Therefore, $\lfloor x \rfloor + 1 \in U(x)$. This shows that $\lceil x \rceil \leq \lfloor x \rfloor + 1$, and $\lceil x \rceil$ is a lower bound for $U(x)$.

Proposition 1.3. *Between any two real numbers there is an irrational number.*

Solution. Suppose $a, b \in \mathbb{R}$ such that $a \leq b$. Let γ be any irrational number. We have $a - \gamma \leq b - \gamma$. We know that there exists $q \in \mathbb{Q}$ such that

$$a - \gamma \leq q \leq b - \gamma.$$

This implies

$$a \leq q + \gamma \leq b.$$

Now, $q + \gamma$ is not a rational number, because if it were, then γ would be a rational number. $\square$

1.2.1 Exercises

Problem 1.1. Suppose U is an [Abelian group](#). Let P be a subset of U with the following properties:

- i. $0 \in P$,
- ii. if $p \in P$ and $-p \in P$ then $p = 0$,
- iii. if $p_1, p_2 \in P$ then $p_1 + p_2 \in P$.

Such a subset is called a *positive cone*. Use P to define a partial order on U .

Conclude (informally) that for an Abelian group, having a partial order is equivalent to having a positive cone.

Problem 1.2. [optional] Given $a, b \in \mathbb{R}$ such that $a < b$, give an algorithm to find a $q \in \mathbb{Q}$ such that $a < q < b$. Argue why your algorithm is correct.

Problem 1.3. Let F be an ordered field. Define the absolute value function $|\cdot|$ on F as follows:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}.$$

Show the following

1. $|x| \geq 0$, and $|x| = 0$ if and only if $x = 0$
2. $|x| = |-x|$
3. $x \leq |x|$
4. $|x + y| \leq |x| + |y|$
5. $|x| \leq a$ if and only if $-a \leq x \leq a$
6. $|x| \geq a$ if and only if $x \geq a$ or $x \leq -a$.

Give a one line reason why the function $d(x, y) = |x - y|$ is not a valid metric on F .

Problem 1.4. Suppose a non-zero polynomial p is expressed as $p(x) = p_m + p_{m-1}x + \cdots + p_0x^m$ for some $m \in \mathbb{N}$, where $p_0 \neq 0$. Let

$$F = \left\{ \frac{p}{q} : p, q \text{ polynomials and } q \neq 0 \right\}.$$

1. Show that F is a field.
2. Let $P = \{p/q \in F : p_0 q_0 \geq 0\}$. Show that P is a positive cone (refer Exercise II, problem 2). Conclude that F is an ordered field.
3. Show that F does not have the Archimedean property.

Problem 1.5. A theorem due to Hölder states that *every Archimedean ordered field is isomorphic^{1.1} to a subfield of $\mathbb{R}$* . Use this to conclude that every Archimedean ordered field can be made into a metric space.

2 Sequences

2.1 Convergent and Cauchy sequences

Assuming the definition of a convergent sequence, let us look at some problems.

Problem 2.1. Let x_n be a sequence and x be a number. Suppose for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \frac{1}{n}$$

1.1. An isomorphism between two ordered fields is a bijection f that preserves the operations and the order, that is, $f(a \cdot b + c) = f(a) \cdot f(b) + f(c)$ and $f(a) \leq f(b)$ whenever $a \leq b$.

for all $n \geq N$. Is x_n convergent? Does this hold for every convergent sequence?

Answer 2.1. Yes, a sequence satisfying the conditions of the problem will be convergent, but not every convergent sequence will satisfy this condition. That is, the condition is sufficient for convergence, but not necessary.

However, one observes that showing that $|x_n - x|$ is eventually dominated by another nonnegative sequence that is going to 0, is sufficient for the convergence of x_n . This technique can often save the hassle of going back to the ε - N definition of convergence.

Proposition 2.1. *Show that a sequence of rational numbers can converge to an irrational number.*

Solution. We have shown earlier that between any two real numbers there is a rational number. Let α be any irrational number. Define x_n as any rational number picked between $\alpha - 1/n$ and α . Then

$$|x_n - \alpha| < \frac{1}{n} \quad \text{for all } n.$$

We have seen that this is a sufficient condition for convergence. □

Proposition 2.2. *Show that a sequence of irrational numbers can converge to a rational number.*

Solution. Consider the sequence $x_n = \sqrt{2}/n$. Clearly, x_n is an irrational number, because if it were not, then $\sqrt{2}$ would be rational. This sequence converges to the limit 0, as the sequence $1/n$ converges to the limit 0. □

Definition 2.1. A sequence $(x_n) \subset F$ is called a *Cauchy sequence*, if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $|x_m - x_n| < \varepsilon$ whenever $m \geq n \geq N$.

Textbooks generally write the condition $m, n \geq N$. But, nothing is lost by assuming $m \geq n$, because $|x_m - x_n| = |x_n - x_m|$. Let us illustrate the definition with a standard example.

Problem 2.2. Show that the sequence $x_n = \frac{1}{n}$ is Cauchy.

Solution. Note that for any $m \geq n$, we have

$$|x_m - x_n| = \frac{m - n}{mn} = \frac{1}{n} \frac{m - n}{m} \leq \frac{1}{n}.$$

Pick any $\varepsilon > 0$. Let $N = \lceil 1/\varepsilon \rceil$. We get

$$\frac{1}{\varepsilon} < N \text{ or } \frac{1}{N} < \varepsilon.$$

So for any $m \geq n \geq N$, we get

$$|x_m - x_n| \leq \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Hence, the sequence is Cauchy. □

Problem 2.3. Let x_n be a sequence such that $|x_{n+1} - x_n| < 2^{-n}$ for all $n \in \mathbb{N}$. Show that x_n is Cauchy.

Solution. For any $m \geq n$, note that

$$\begin{aligned} |x_m - x_n| &= |x_m - x_{m-1} + x_{m-1} - x_{m-2} + \cdots + x_{n+1} - x_n| \\ &\leq |x_m - x_{m-1}| + \cdots + |x_{n+1} - x_n| \\ &\leq 2^{-n} + \cdots + 2^{-(m-1)} \\ &= 2^{-(n-1)}[2^{-1} + \cdots + 2^{-(m-n)}] \\ &\leq 2^{-(n-1)}. \end{aligned}$$

Pick any $\varepsilon > 0$. As the sequence $2^{-(n-1)}$ is convergent with the limit 0, there exists $N \in \mathbb{N}$ such that

$$2^{-(n-1)} < \varepsilon \quad \text{for all } n \geq N.$$

Therefore, for any $m \geq n \geq N$, we get

$$|x_m - x_n| \leq 2^{-(n-1)} < \varepsilon. \quad \square$$

From this problem we extract the following proposition.

Proposition 2.3. Suppose s_n is a sequence such that the series $\sum s_n$ converges absolutely. Let x_n be another sequence such that $|x_n - x_{n-1}| \leq |s_n|$ for all $n > 1$. Show that x_n is a Cauchy sequence.

Solution. Define $S_n = \sum_{k=1}^n |s_k|$. The series $\sum s_n$ converging absolutely is the same as the sequence S_n converging. Therefore, S_n is also a Cauchy sequence. Now, note that for any $m > n$, we have

$$\begin{aligned} |x_m - x_n| &= |x_m - x_{m-1} + x_{m-1} - x_{m-2} + \cdots + x_{n+1} - x_n| \\ &\leq |x_m - x_{m-1}| + \cdots + |x_{n+1} - x_n| \\ &\leq |s_{m-1}| + \cdots + |s_n| \\ &= S_m - S_n \\ &\leq |S_m - S_n|. \end{aligned}$$

Choose any $\varepsilon > 0$. Since S_n is Cauchy, there exists $N \in \mathbb{N}$ such that

$$|S_m - S_n| < \varepsilon \quad \text{for all } m, n \geq N.$$

Therefore, whenever $m, n \geq N$, we also have

$$|x_m - x_n| \leq |S_m - S_n| < \varepsilon. \quad \square$$

Note that absolute convergence of $\sum s_n$ is necessary. The series $\sum s_n$ where $s_n = 1/n(-1)^{n+1}$ is convergent but not absolutely convergent. The sequence of partial sums $S_n = \sum |s_n|$ satisfies $|S_n - S_{n-1}| = 1/n = |s_n|$. However, the sequence is not convergent.

Now, to take our understanding further, let us study a special sequence. Let

$$e_n = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}.$$

Note that $j! \geq 2^j$ for all $j \geq 4$. To see this we note that $4! \geq 2^4$, and from here we see inductively that

$$(j+1)! = (j+1)j! \geq (j+1)2^j \geq 2 \cdot 2^j = 2^{j+1}.$$

Clearly, e_n is a monotonically increasing sequence. And the above relation gives

$$e_n \leq 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{2^4} + \cdots + \frac{1}{2^n} \leq 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{8} \leq 3.$$

We conclude that e_n is convergent. The famous number e is (often) *defined* as the limit of this sequence, that is,

$$e = \lim_{n \rightarrow \infty} e_n = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \cdots.$$

Problem 2.4. Show that $\lim \left(1 + \frac{1}{n}\right)^n = e$.

Solution. Let $x_n = \left(1 + \frac{1}{n}\right)^n$. Use the binomial theorem to get

$$x_n = \sum_{j=0}^n \frac{p_{n,j}}{j!} \quad \text{where} \quad p_{n,j} = \prod_{k=0}^{j-1} \left(1 - \frac{k}{n}\right),$$

with the understanding that $p_{n,0} = 1$. As an aside, since $(1 - k/n)$ is increasing in n , so is $p_{n,j}$ and, therefore, x_n . Further, we have $0 \leq p_{n,j} \leq 1$. This gives

$$x_n \leq e_n < e.$$

Being monotonically increasing and bounded above, x_n is also convergent. One may intuit that e is the limit of x_n , noting that $\lim_{n \rightarrow \infty} p_{n,j} = 1$. However, the problem is that the limit cannot be exchanged with the sum, as number of terms in the sum itself grows with n . To circumvent this, we define a new function

$$P_n(m) = \sum_{j=0}^m \frac{p_{n,j}}{j!}.$$

Note that $x_n = P_n(n)$. Further note that for all $n \geq m$, we have

$$|x_n - e| \leq |x_n - P_n(m)| + |P_n(m) - e|.$$

Now, choose any $\varepsilon > 0$. Since e_n converges to the limit e , there exists an m_0 such that

$$e_{m_0} > e - \frac{\varepsilon}{2}.$$

Indeed the inequality will hold for all $n \geq m_0$, but we focus only the term e_{m_0} . Further, since $p_{n,j}$ goes to 1 as n grows to infinity, we get that $P_n(m_0)$ converges to the limit e_{m_0} . Since m_0 is independent of n , this time the limit can easily be taken inside the sum. Due to this convergence, we know that there exists an $N \geq m_0$ such that

$$P_N(m_0) \geq e_{m_0} - \frac{\varepsilon}{2}.$$

Now we note that for all $n \geq N \geq m_0$, since x_n is an increasing function, we get

$$e > x_n \geq x_N = P_N(N) \geq P_N(m_0) \geq e_{m_0} - \frac{\varepsilon}{2} \geq e - \varepsilon.$$

The inequality $P_N(N) \geq P_N(m_0)$ is simply the consequence of truncating the sum $P_N(N)$ of N nonnegative numbers to just m_0 of them. The extreme sides of the obtained inequality gives

$$e - \varepsilon \leq x_n < e \quad \text{for all } n \geq N.$$

This shows that the sequence x_n is convergent with the limit e . □

2.2 Limit superior and limit inferior

Let x_n be a sequence. The number L called the *limit superior* of x_n , denoted by $\limsup x_n$ or $\overline{\lim} x_n$, are defined in either of the following ways.

(I) For every $\varepsilon > 0$

- (i) only a finitely many terms of x_n are larger than $L + \varepsilon$ and
- (ii) infinitely many terms of x_n are strictly larger than $L - \varepsilon$.

Similarly, the *limit inferior* of x_n is a number L , denoted by $\liminf x_n$ or $\underline{\lim} x_n$, such that for every $\varepsilon > 0$

(i) only a finitely many terms of x_n are smaller than $L - \varepsilon$ and

(ii) infinitely many terms of x_n are strictly smaller than $L + \varepsilon$.

(II) Let $X_k = \{x_k, x_{k+1}, \dots\}$. Then, alternatively, we have

$$\limsup x_n = \lim_{k \rightarrow \infty} \sup X_k \text{ and } \liminf x_n = \lim_{k \rightarrow \infty} \inf X_k.$$

It is assumed here that the underlying space has the LUB property, otherwise the infimum and supremum may not make sense. Unless mentioned otherwise, for this discussion, all sequences will be in $\mathbb{R}$. Now, note that $\sup X_k$ and $\inf X_k$ are decreasing and increasing sequences respectively. Therefore, if we include $\pm\infty$ in our system, then every sequence x_n has a limit superior and a limit inferior.

(III) Let $\Lambda(x_n)$ be the set of all subsequential limits of x_n . That is, for every $l \in \Lambda(x_n)$, there exists a subsequence x_{n_j} of x_n such that x_{n_j} converges to l . Then, another definition of limit superior and limit inferior is as follows:

$$\limsup x_n = \sup \Lambda(x_n) \text{ and } \liminf x_n = \inf \Lambda(x_n).$$

Problem 2.5. Let x_n and y_n be two sequences. Is it true that

$$\limsup x_n y_n \leq (\limsup x_n)(\limsup y_n)?$$

Solution. Not necessarily. Consider the sequence

$$0, -1, 0, -1, \dots$$

Let x_n and y_n be both this sequence. Then $\limsup x_n = 0$. So the right hand side of the inequality is 0. The sequence $x_n y_n$ is

$$0, 1, 0, 1, \dots$$

Then the left hand side $\limsup x_n y_n = 1$. This is larger than the right hand side. $\square$

Question 2.1. Let q_n be any enumeration of the rational numbers. What is $\Lambda(q_n)$? Conclude that a sequence may not only have a countably infinite set of subsequential limits, but also an uncountable one.

The following properties of limit superior and limit inferior have been seen.

1. $\liminf x_n \leq \limsup x_n$ for all x_n .
2. If x_n is convergent, then $\limsup x_n = \liminf x_n = \lim x_n$.

3. A corollary of the point above is that every Cauchy sequence in $\mathbb{R}$ is convergent (earlier we had seen that every convergent sequence is Cauchy).
4. There is a subsequence of x_n that converges to $\limsup x_n$ and $\liminf x_n$. As a corollary we get Bolzano-Weierstrass theorem.
5. If $x_n \neq 0$, then

$$\limsup |x_n|^{\frac{1}{n}} \leq \limsup \left| \frac{x_{n+1}}{x_n} \right|.$$

For $\liminf$, the inequality goes in the other direction. This also tells us that if $|x_{n+1}/x_n|$ is convergent, then so is $|x_n|^{1/n}$, and to the same limit.

2.2.1 Exercises

Problem 2.6. Show that the definitions (I), (II) and (III) are indeed equivalent.

Problem 2.7. Show that the supremum and the infimum in definition (III) can be replaced by maximum and minimum, respectively.

Problem 2.8. Let $x_n = \frac{(-1)^n n}{(-1)^{n+1} + n}$. Calculate $\limsup x_n$ and $\liminf x_n$.

Problem 2.9. Let x_n be a sequence such that the subsequence x_{2n} is convergent with the limit L , and the subsequence x_{2n-1} is also convergent with the same limit L . Show that x_n is convergent with the limit L .

Problem 2.10. Let x_n be any convergent sequence. Define a new sequence σ_n where

$$\sigma_n = \frac{(x_1 + \cdots + x_n)}{n}.$$

Show that σ_n is also convergent and $\lim \sigma_n = \lim x_n$.

For the statements in coming problems either give a proof or disprove them with a clear counterexample.

Problem 2.11. Let x_n be a bounded sequence. Then $\sup \{x_n : n \in \mathbb{N}\} = \limsup x_n$.

Problem 2.12. For every sequence x_n we have $\limsup x_n = -\limsup (-x_n)$.

Problem 2.13. For every sequence x_n we have $\liminf x_n = -\limsup (-x_n)$.

Problem 2.14. Let x_n be a sequence and L be a number. Suppose for every $\varepsilon > 0$

- i. there exists an $N \in \mathbb{N}$ such that $x_n < L + \varepsilon$ for all $n \geq N$,
- ii. there exists an $m \in \mathbb{N}$ such that $x_m > L - \varepsilon$.

Then L must be $\limsup x_n$.

Problem 2.15. Let x_n be a sequence of **distinct** rational numbers. Then

$$\limsup \sin x_n = 1.$$

Problem 2.16. Suppose x_n is a bounded sequence such that it is **not** convergent. Then x_n has at least 2 subsequential limits.

3 Series

3.1 Convergence, absolute and conditional convergence

A series is an infinite sum of the terms of a sequence, usually denoted by $\sum x_n$. In the notation, the sum is just a formal symbol, and does not necessarily represent the sum of the series. When we say two series $\sum x_n$ and $\sum y_n$ are equal, we do not mean that their sum is equal (for the sum may not exist), but just that the sequence x_n and y_n are identical.

For any series $\sum x_n$ one can construct the sequence of its partial sums

$$s_m = \sum_{n=1}^m x_n.$$

Thus, one can talk about the convergence of a series by studying the convergence of the sequence of its partial sums. A series is called *convergent* if the sequence s_m of its partial sums is convergent, *absolutely convergent* if the positive-termed series $\sum |x_n|$ is convergent, and *conditionally convergent* if the series is convergent but not absolutely convergent.

One may be able to decide whether a series converges using tests like the comparison test, root test, ratio test and integral test.

Proposition 3.1. *If a series $\sum x_n$ is absolutely convergent then it is convergent.*

Solution. Trivially we have $x_n \leq |x_n|$. So the series converges by comparison test. To elaborate, for any $m \geq n$ we have

$$\left| \sum_{k=n}^m x_k \right| \leq \sum_{k=n}^m |x_k|.$$

Sine $\sum |x_n|$ is convergent, the right hand side can eventually be made arbitrarily small. So the series $\sum x_n$ converges by Cauchy criterion. $\square$

Proposition 3.2. *Every rearrangement of an absolutely convergent series has the same sum.*

Solution. Let L be the sum of the series $\sum a_n$ and let $\sum b_n$ be any rearrangement of it. Then there exists a bijection f of $\mathbb{N}$ such that $a_n = b_{f(n)}$ for all n .

Choose any $\varepsilon > 0$. Since $\sum a_n$ is absolutely convergent, there exists a natural number N such that

$$\sum_{n=N+1}^{\infty} |a_n| < \frac{\varepsilon}{2}.$$

Let s_m and t_m be the sequence of partial sums of the series $\sum a_n$ and $\sum b_n$ respectively. Let $K = \max \{f(1), \dots, f(N)\}$. For any $m \geq K$, consider the difference

$$|t_m - s_m| = \left| \sum_{n=1}^M b_n - \sum_{n=1}^M a_n \right|.$$

The terms in both the partial sums come from the set $\{a_k\}$. Any term a_k either appears in both the partial sums, in which case it gets cancelled, or it appears in at most one of the two partial sums. By our choice of m , the terms $a_1, \dots, a_N$ are among those that appear in both and get cancelled. Therefore we conclude that

$$|t_m - s_m| < \sum_{n=N+1}^{\infty} |a_n| < \frac{\varepsilon}{2}.$$

Finally, since s_m is convergent to the limit L , we can pick $m \geq K$ such that

$$|s_m - L| < \frac{\varepsilon}{2} \quad \text{for all } m \geq K.$$

Finally, for all $m \geq K$ we get

$$|t_m - L| \leq |t_m - s_m| + |s_m - L| < \varepsilon. \quad \square$$

Proposition 3.3. [Riemann rearrangement theorem] *If the series $\sum x_n$ is conditionally convergent, then given any $\alpha \in \mathbb{R}$, there exists a rearrangement $\sum y_n$ of the series $\sum x_n$, such that $\sum y_n$ is convergent with the sum α .*

Solution. *A translation of Riemann's original proof of the series rearrangement theorem: a conditionally convergent series can be rearranged to have any arbitrary sum.*

... infinite series fall into two distinct classes, depending on whether or not they remain convergent when all the terms are made positive. In the first class the terms can be arbitrarily rearranged; in the second, on the other hand, the value is dependent on the ordering of the terms.

Indeed, if we denote the positive terms of a series in the second class by $a_1, a_2, a_3, \dots$ and the negative terms by $-b_1, -b_2, -b_3, \dots$ then it is clear that $\sum a_n$ as well as $\sum b_n$ must be infinite. For if they were both finite, the series would still be convergent after making all the signs the same. If only one were infinite, then the series would diverge.

Clearly now an arbitrarily given value α can be obtained by a suitable reordering of the terms. We take alternately the positive terms of the series until the sum is greater than α , and then the negative terms until the sum is less than α . The deviation from α never amounts to more than the size of the term at the last place the signs were switched. Now, since the numbers a_n as well as the numbers b_n become infinitely small with increasing index, so also are the deviations from α . If we proceed sufficiently far in the series, the deviation becomes arbitrarily small, that is, the series converges to α . $\square$

Proposition 3.4. *If $\sum a_n$ and $\sum b_n$ are absolutely convergent, then so is $\sum a_n b_n$.*

Solution. Since $\sum b_n$ is absolutely convergent, $|b_n|$ goes to 0. In particular $|b_n|$ is bounded, that is, there exists M such that $|b_n| \leq M$ for all n . So,

$$\sum_{n=1}^{\infty} |a_n b_n| \leq M \sum_{n=1}^{\infty} |a_n|.$$

The partial sums of $\sum |a_n b_n|$ are increasing and bounded, and hence the sum converges. Hence $\sum a_n b_n$ converges absolutely. $\square$

Let $\sum a_n$ and $\sum b_n$ be two series indexed from 0 (instead of 1). We define a new series $\sum c_n$ where

$$c_n = \sum_{k=0}^n a_k b_{n-k}.$$

The series $\sum c_n$ is called the *Cauchy product* of $\sum a_n$ and $\sum b_n$.

Proposition 3.5. *The Cauchy product of two absolutely convergent series is absolutely convergent.*

Solution. Let $\sum a_n$ and $\sum b_n$ be absolutely convergent. Let $\sum c_n$ be their Cauchy product. Then

$$\sum_{n=0}^N |c_n| \leq \sum_{n=0}^N \sum_{k=0}^n |a_k| |b_{n-k}| \leq \sum_{n=0}^N |a_n| \sum_{n=0}^N |b_n| \leq \sum_{n=0}^{\infty} |a_n| \sum_{n=0}^{\infty} |b_n|,$$

where the second inequality relies on the fact that the terms of the sum on the left hand of it are contained in the terms of the sum on the right, and no term is negative. Again, the partial sums are increasing and bounded, and hence the series is convergent. $\square$

3.1.1 Exercises

Either prove the following statements or disprove with a counterexample.

Problem 3.1. If the series $\sum a_n$ and $\sum b_n$ are convergent, then so is the series $\sum a_n b_n$.

Problem 3.2. The Cauchy product of two convergent series is convergent.